Battery State-of-Charge and Remaining Charge-Time Estimation from Li-ion Drive-Cycle Data

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Problem and objectives

- SOC cannot be measured directly - it must be estimated from voltage, current and temperature
- Goal 1: estimate SOC (%) of a Li-ion cell
- Goal 2: estimate remaining charging time in CC-CV charge segments
- Goal 3: choose the final method on validation evidence, robustness and simplicity
- Goal 4: deliver a frozen, reproducible software package

Data and frozen split

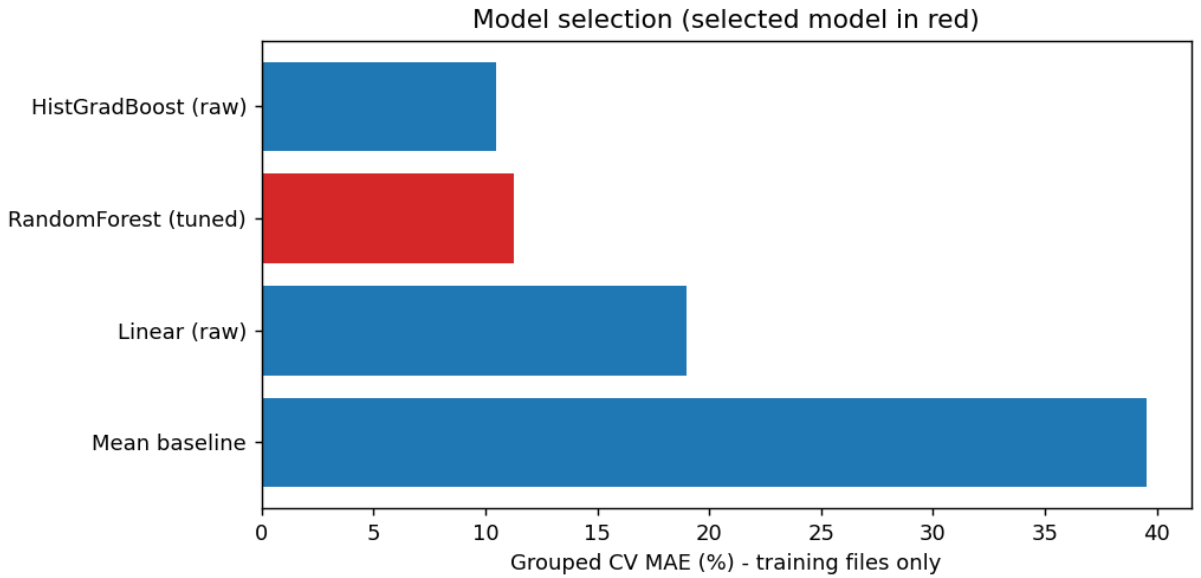
- 5,356 rows in 23 files
- Features: voltage_v, current_a, cell_temp_c, power_w, ambient_temp_c
- Split by file_id: 18 train / 5 held-out files - no leakage
- Split frozen; held-out files used for the final score only
- Charging target = seconds to the end of each CHA segment

Methodology

- Candidates: mean, linear, random forest, histogram gradient boosting (MLP rejected)
- Grouped cross-validation on training files only
- Selection: GroupKFold CV on train files; within 10% of best CV MAE -> simplest; worst fold <= 2x mean
- Metrics: MAE, RMSE, R2, worst-file MAE, p95 error; SOC clipped to 0-100 %
- Engineered features excluded - they did not beat raw features

Model selection evidence

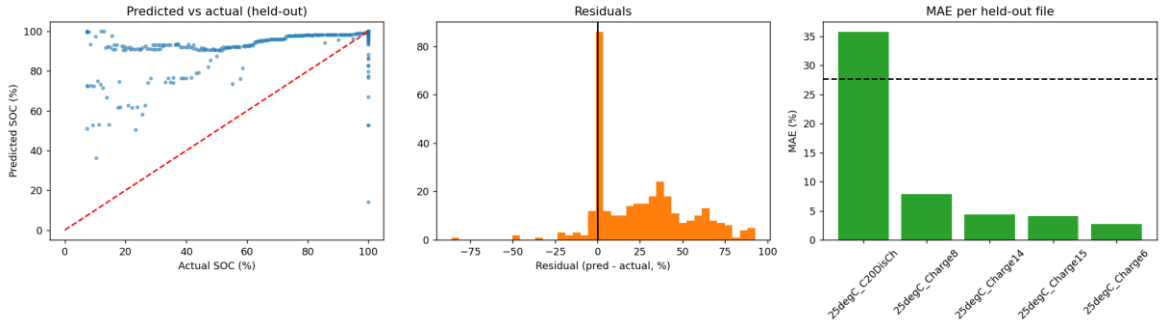
model	rank	cv_MAE	cv_RMSE	cv_R2	worst_fold_MAE
Mean baseline	0	39.551	41.769	-8.591	44.118
Linear (raw)	1	18.972	29.117	-8.728	29.847
RandomForest (tuned)	2	11.292	20.415	-1.312	26.332
HistGradBoost (raw)	3	10.462	19.363	-1.369	24.125



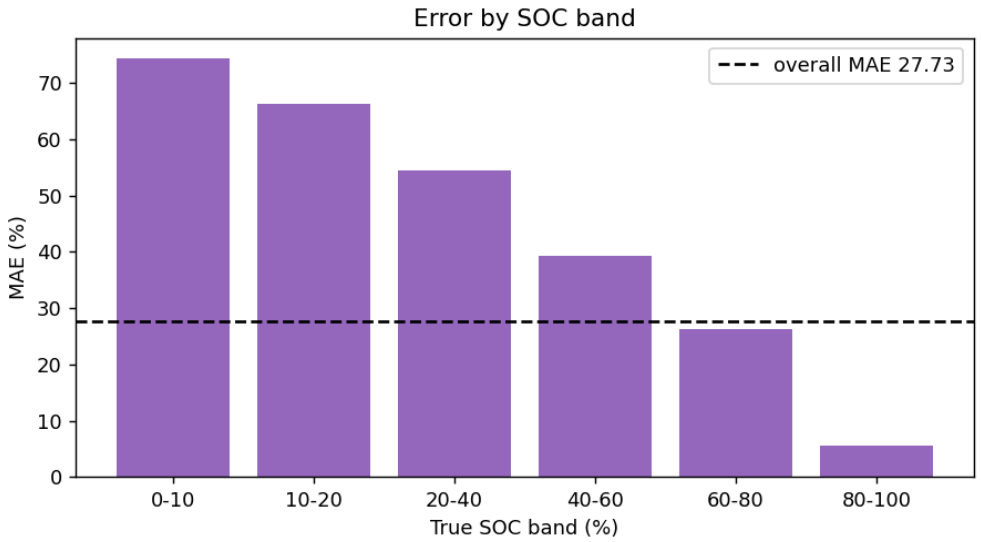
Final results on held-out files

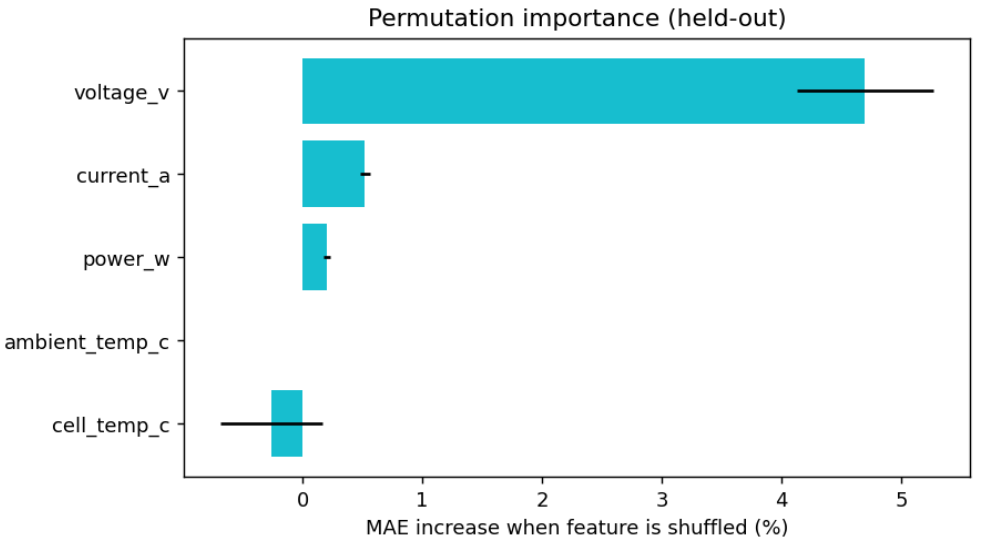
model	MAE	RMSE	R2
Mean baseline	28.928	32.192	-0.033
Linear regression (raw)	18.104	25.163	0.369
RandomForest (tuned) (final)	27.725	37.708	-0.417

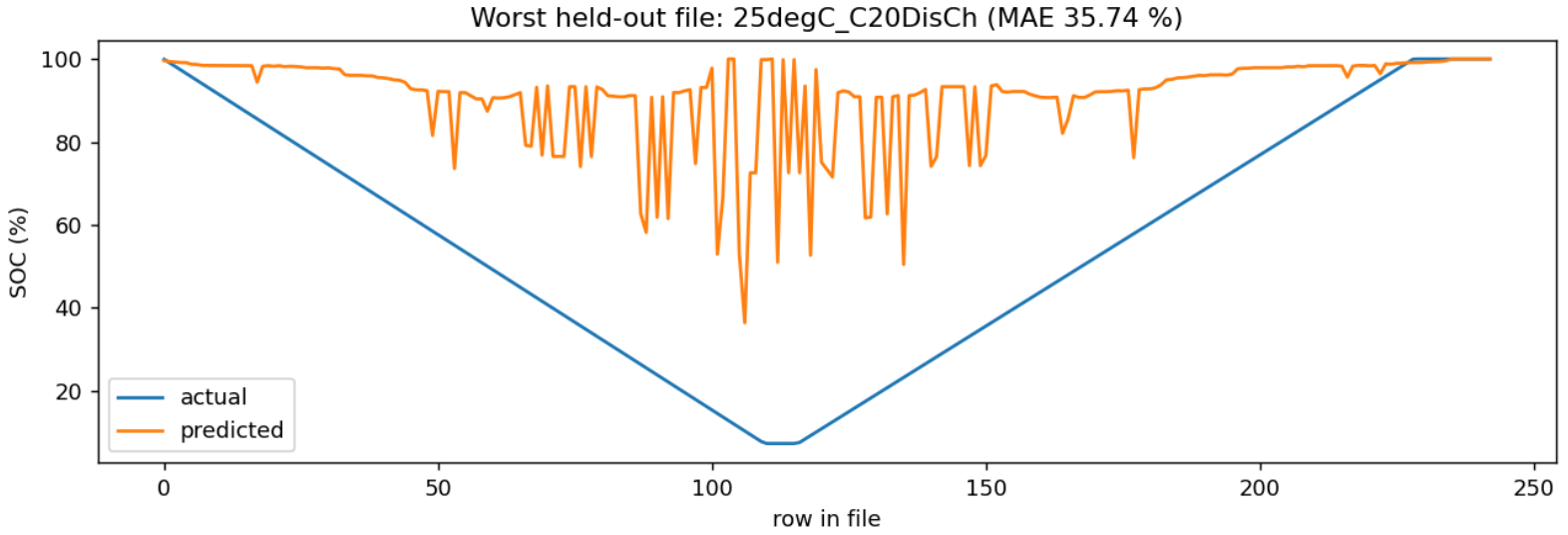
- SOC: MAE 27.73 %, RMSE 37.71 %, R2 -0.417
- Worst file MAE 35.74 %; p95 |err| 74.80 %
- Charge time: MAE 26515 s, R2 -1.27 (5 file(s) - indicative)



Error analysis







Limitations and failure conditions

- Small held-out set - only 5 files / 328 rows, so every metric has wide uncertainty.
- Uneven error across SOC - the highest MAE is in the 0-10 % band (74.33 % vs overall 27.73 %).
- Occasional large errors - RMSE (37.71 %) is 1.4x the MAE; the worst file has MAE 35.74 %.
- Robustness flag from Day 22 - the worst cross-validation fold MAE exceeded 2x the mean fold MAE, so accuracy varies strongly between files.
- Limited operating conditions - ambient temperature in the data: 25 C; other temperatures are untested.
- Static framing - each row is predicted independently (no history), so sensor noise or drift is not filtered.

Reproducible software package

- src/soc_pipeline: pipeline module + train / evaluate / predict command line
- models: frozen joblib pipelines + frozen_config.json + cross-validation results
- tests: pytest suite; outputs: example input -> output; figures + tables for every claim
- python scripts/predict.py --model-dir models --input in.csv --output out.csv
- Retraining from the frozen estimator + seed is checked against the stored model

Future scope and conclusion

- Investigate why the history-based engineered features failed to beat raw features and repair them.
- Add sequence-aware models (windowed features, GRU / LSTM) and compare them under the same grouped protocol.
- Collect data at more ambient temperatures and drive cycles; use leave-one-temperature-out validation.
- Add uncertainty estimates and an out-of-range input detector so unreliable predictions are flagged.
- Collect more charging segments before treating charge time as a production deliverable.
- Conclusion: A frozen, reproducible pipeline estimates SOC with MAE 27.73 % on unseen files. The main risks are the small number of held-out files, the static (history-free) framing and untested conditions;...

THANK YOU